Trip/Training Report
Multiple Cropping Center, Chiang Mai University (MCC-CMU), Thailand
14-25 October 2002

Key resource persons/trainers: Dr. Suan Pheng Kam / IRRI, Dr. Christophe Le Page / Tera-Cirad, France, and Dr. Olivier Barreteau / *Cemagref*, France.

IRRI Staff participants: François Bousquet, Guy Trébuil, Suan Pheng Kam.

Local organizers : Dr. Methi Ekasingh and Dr. Benchaphun Shinawatra-Ekasingh, MCC-CMU. The course was opened by Dr. Pittaya Suamsiri, Dean of the Faculty of agriculture, CMU.

Group photo

Purpose
This two-week course corresponded to steps 5 and 6 (out of 12) of the on-going Asia IT&C Project on Multi-Agent Systems, Social Sciences and Integrated Natural Resource Management (INRM) coordinated by the IRRI-Cirad team based at IRRI Thailand office (see a project description sheet in appendix 3).
Also supported by the Regional Cooperation Office of the Embassy of France in Bangkok, IRRI, and Cirad, the sessions aimed at improving the participants understanding of how to link MAS with GIS when developing applications in community-based approaches to participatory Integrated Watershed Management.

Course objectives
(i) To improve the participants' understanding of the complementarities between GIS and MAS,
(ii) To demonstrate how to link GIS outputs with MAS models,
(iii) To introduce the principles and tools for integrated watershed management, with an emphasis on multi-stakeholders decision-making processes, and
(iv) To support the participants in developing their personal applications.

Summary of the training
The regional training course on Multi-Agent Systems (MAS) and Geographic Information Systems (GIS) for Integrated Watershed Management (IWM) was held in northern Thailand at the Multiple Cropping Center, Chiang Mai University (MCC-CMU) on 14-25 October 2002. As usual, the local organizing committee did an outstanding job and created an excellent working atmosphere (see evaluation results for course features in appendix 4).
The first week of the course focused on how to link outputs from GIS with MAS. After introducing and illustrating the main principles of GIS, Dr. S.P. Kam led sessions to demonstrate how to convert GIS files to be able to use them in MAS. Dr. Christophe Le Page presented MAS, with an emphasis on their use of spatial entities and their management by models. Hands-on exercices on how to link MAS with GIS were performed, and two case studies were presented. One case study, on Mae Salaep watershed (North Thailand), was used as a cross-cutting exercise (GIS? MAS, Role game).
The second week focused on the use of agent-based modeling for integrated watershed management. After introducing the principles of IWM, Dr. Olivier Barreteau lectured about

the different tools which can be used, with an emphasis on those helping to elucidate decision-making processes. This second part of the course was also illustrated by three case studies.
During this course, the usefulness of three complementary types of UML (Unified Modeling Language) diagrams for organizing an interdisciplinary preparation of multi-agent models was demonstrated. The complementary roles of such models and role-playing games in IWM was also illustrated through two case studies. The results of the course evaluation show that these two topics were well-received by the participants, as well as the time and trainers' support provided to small groups working on their own applications.

The participants
Eighteen trainees from eight countries (Thailand, the Philippines, Vietnam, Indonesia, Buthan, Bangladesh, Australia and Japan), and six observers (from Thailand, France, Germany and Japan) attended this two-week training course (see the list of participants in appendix 2).
Beyond the regular participants (corresponding to one half of the group) who attended the previous sessions, newcomers continue to join in the project. This heterogeneity among trainees necessitates adjustments of the contents and presentations to ensure that the course benefits the whole group.

Training approach and structure of the course
Every day, an initial 1h30 lecture was followed by hands-on exercices in the computer laboratory.
Generally, afternoons started with presentations and discussions of case studies, followed by either more practical exercices or trainees work on their personal or group applications.
This last activity was very highly rated in the course evaluation by the participants. As usual, the last morning of the course was fully allocated to eight trainees' presentations and their discussions.

Course outlines
See the detailed schedule of the successive sessions in appendix 1.

Course instructors and resource persons
- Dr. Christophe Le Page (ecologist and modeller) : lectures on linking GIS with MAS, presentation of several case studies, guided several hands-on exercices, and supported trainees work on their personal projects,
- Dr. Suan Pheng Kam (GIS specialist): lectures on introduction to GIS, practical exercices on GIS, and conversion of GIS files for MAS,
- Dr. Olivier Barreteau (water management and modeller): lectures on integrated watershed management and MAS, presentation of several case studies, provided support during hands-on exercices, and supported trainees work on their personal projects,
- François Bousquet (IRRI-Cirad, ITO): project coordinator, presented several case studies, led several hands-on exercices, and supported trainees work on their personal projects,
- Guy Trébuil (IRRI-Cirad, ITO): resource person, presented case studies, and supported trainees work on their personal projects,
- Wipawan Boonsom, secretary, IRRI-Cirad Project at ITO.

Training materials
Every participant in this course received the following documents:

(i) A "Manual" entitled MAS, GIS, and Integrated Watershed Management reproducing all the slides used by the trainers during the two weeks,
(ii) A second volume presenting the hands-on exercices and an up-to-date copy of the CORMAS tutorial (as of September 2002),
(iii) A third volume made of six reference papers,
(iv) A CD providing all the files related to the lectures (ppt presentations), the hands-on exercices, and the MAS softwares used during the training (CORMAS and Visual Works 7).

Trainees' presentations
A total of ten short presentations were made by the participants on the following topics:

- Integrating natural resources and social dynamics: case study of Phataen national park, Ubon Ratchathani, Thailand (Chulalongkorn University),
- Managing forest plantations using the "teakwood market game" in Eastern Java (CIFOR),
- Diffusion of a new rice technology in Vietnam (NISF and Cantho University),
- Model of tree and crop adoption insmallholder farms in uplands of Claveria, Msamis Oriental, the Philippines (UPLB and ICRAF),
- Irrigation water management in Lingmutheychu watershed, Bhutan (Department of research and development services, MoA, and RNR-RC Bajo)
- Multi-agent systems for integrated watershed management in Mae Hae, Chiang Mai Province, northern Thailand (MCC-CMU),
- Understanding land-use changes in small watersheds of upper northeast Thailand (JIRCAS and Land Development Department),
- MAS assisted interactive policy design for coastal resource management in Bohol, the Philippines (UP at Diliman campus),
- Modelling land-use changes in Nong Chok, Pathum Thani Province, Central Thailand (School of advanced technologies, Asian Institute of Technology),
- Modelling groundwater management in south Tarawa (Kiribati) with MAS (Australian National University and Cirad).

Following each presentation, its contents was discussed by the participants and trainers provided comments. This sharing of personal experiences was considered as a strength of the course during its evaluation (see appendix 4).

Course evaluation
Twenty participants returned the evaluation forms. The course was rated as "excellent" by 67% of the respondants, and as "good" by the remaining ones (see details in apprendix 4). According to the respondants, the course objectives related to improvements in understanding new concepts, approaches, methods, and tools to use MAS with GIS for IWM were fully attained. And only 5% of the participants were not able to make progress in developing their personal applications during this course.
The contents of all the topics presented during the course were found relevant, useful, and well-presented. Several case study presentations could have been shorter. The usefulness of the UML diagrams and group work on personal projects obtained the highest marks. The contents of the role-playing games was considered as excellent, but several respondants found that not enough time was allocated to these activities. The same remark can be said for the UML diagrams and the conversion of GIS files for MAS.

The following strengths of this course were mentioned by the trainees (based on a total of 48 answers provided by 20 participants):

- General organization of the course, good mix of complementary activities, the group dynamics,
- Up-to-date contents of the course, and applicability of tools for interdisciplinary work,
- Quality of trainers and easy access to trainers,
- Group work on own projects supported by trainers, presentations to share experiences,
- Usefulness of Unified Modelling Language (UML) for interdisciplinary conception of models,
- Complementary functions of role-playing games and agent-based models.

Only a few weaknesses were mentioned in the evaluation forms (total of 12 answers from 20 participants), and the following remarks were recorded:

- The course was too short: not enough time for computer exercices, for mastering UML diagrams, for practicing on GIS-MAS linkages, for learning or improving knowledge on “smalltalk” programming language under the CORMAS simulation platform,
- Opinions were split regarding the number and amount of time allocated to the presentation of case studies (not all were find very relevant to this course),

Only two suggestions were made regarding possible topics to be added in this course: the study of IWM with multiple conflicts among actors, and the use of the proposed approaches at larger scales for meso or macro watersheds.
The organizers and trainers responded by providing information on a new one-week course on UML and CORMAS offered in France which is complementary to the training sessions organized by the current Asia IT&C Project.

Progress made
One year after initiating this process, an active network of several university teams developing applications and models to address concrete INRM problems in their regions is now functioning.

The models being developed by these teams help understanding the resilience of agroecosystems, the effects of different options for managing resources, and the role of social organizations for regulating landscape dynamics. Based on these preliminary results, it seems possible to plan the preparation of a collective publication of this young network presenting their first applications in early 2004.

In his address to the participants, Dean Pittaya mentioned the usefulness of these approach and tools particularly to manage the uplands. He also confirmed the wish of his faculty to help promote such activites in the Mekong sub-region in the future by relying on this new network.

Several members of this network of MAS users for INRM in Southeast Asia are now starting doctoral studies in this field in different countries such as Canada, France, and Japan. During this training course, several other participants confirmed their wish to do the same. While another one decided to include his new knowledge and the case study he started to develop during this training into his research for an M.Sc. degree at MCC-CMU.

The next meeting of the network members is planned in late March 2003 at Chulalongkorn University in Bangkok. This one-week workshop will focus on MAS and environmental issues, and will be led by a resource person from Manchester Metropolitan University, UK.

Appendices

1. Full schedule of the course
2. List of course participants
3. Asia IT&C project description sheet
4. Results of the course evaluation by the participants.

Appendix 1. Schedule of the International Training on Multi-Agent Systems, Geographic Information Systems and Integrated Watershed Management At Nakorn Na Lampang Conference Room, Multiple Cropping Center, Chiang Mai University, 14-25th October, 2002.

Day	*8h30-10h*	*10h30-12h*	*13h30-15h*	*15h30-17h*	*Evening*
Monday 14	Registration, Opening ceremony, Introduction of Participants	Introductory Conference	"Shadoc" role-playing game		Reception Dinner at MCC (17:00)
Tuesday 15	Introduction to MAS	Exercice on MAS	Introduction to GIS (1)	Exercice on GIS (1)	
Wednesday 16	Introduction to GIS (2)	Exercice on GIS (2)	Discussions with trainers, and hands-on exercices		
Thursday 17	Spatialized MAS	Exercice on UML	MAS-GIS Watershed Management case study	Exercice on MAS and GIS	
Friday 18	Integrated Watershed Management	Exercice on MAS	MAS-GIS Water management case study	Exercice on MAS and GIS	
Saturday 19	Field Trip to Queen Sirikit Botanical Garden and Ban Tawai, Chiang Mai				
Monday 21	Tools for IWM	Exercices on case study	MAS	"Mae Salaep" Role game	
Tuesday 22	MAS and decision-making processes	Exercices on case study	MAS	Exercices on case study	
Wednesday 23	IWM examples	Exercices on case study	Discussions with trainers, hands-on exercices		
Thursday 24	Presentation of the "Firma" Project	Exercices on case study	Exercices on case study		Farewell Diner
Friday 25	Participants' presentations to the group		Evaluation, Discussion & Closing Ceremony		

Appendix 2. List of participants in the training course on Multi-Agent Systems, Geographic Information Systems, and Integrated Watershed Management at the Multiple Cropping Center, Chiang Mai University, 14-25th October, 2002.

Name	Mailing Address	Phone/ Fax
TRAINERS/ORGANIZERS		
Dr. Benchaphun Ekasingh	Multiple Cropping Center, Chiang Mai University 239 Huay Kaew Road, Chiang Mai 50200, Thailand	Tel. 6653-22127 Fax. 6653-21000
Dr. Christophe Le Page	Cirad TERA, Avenue Agropolis, 34398 Montpellier Cedex 5, France	Tel. 33-4675938 Fax. 33-4675938
Dr. François Bousquet	IRRI Thailand Office, P.O. Box 9-159 Chatuchak, Bangkok 10900	Tel. 662-942702 Fax. 662-56148
Dr. Guy Trébuil	IRRI Thailand Office, P.O. Box 9-159 Chatuchak, Bangkok 10900	Tel. 662-942702 Fax. 662-56148
Dr. Methi Ekasingh	Multiple Cropping Center, Chiang Mai University 239 Huay Kaew Road, Chiang Mai 50200, Thailand	Tel. 6653-22127 Fax. 6653-21000
Dr. Olivier Barreteau	UR Irrigation, *Cemagref* B.P. 5095 34033 Montpellier, France	Tel. 33-4670463 Fax. 33-4676357
Dr. Suan Pheng Kam	IRRI DAPO Box 7777	Tel. 632-845056
	Metro Manila , the Philippines	ext. 592 or 627 Fax. 632-845060
TRAINEES		
Ms. Anne Dray	11 Rue Lalanne, 31100 Toulouse, France	
Dr. Chanchai Saengchyoswat	Multiple Cropping Center, Chiang Mai University 239 Huay Kaew Road, Chiang Mai 50200, Thailand	Tel. 6653-22127 Fax. 6653-21000
Dr. Damasa M. Macandog	Institute of Biological Sciences, University of Philippines at Los Banos, College Laguna 4031, the Philippines	Tel. 63-4953674
Dr. Kenji Suzuki	Japan International Research Center for Agricultural Sciences 1-1 Ohwashi, Tsukuba, Ibaraki 3058686, Japan	Tel. 81-2983863 Fax 81-2983863
Mr. Le Canh Dung	Farming Systems Institute Cantho University, Vietnam	Tel. 8471-83126
Mr. Marc Delgado	ECO INFO Lab, Institute of Biological Sciences University of Philippines at Los Banos College Laguna 4031, the Philippines	Tel. 6349536741
Ms. Mayura Prabpan	Soil Survey Laboratory, ITCAD, Mitrapap Road Amphoe Muang Khon Kaen 40000, Thailand	Tel. 6643-24675
Dr. Nantana Gajaseni	Biology Department, Faculty of Science Chulalongkorn University, Bangkok 10330, Thailand	Tel. 662-218536 Fax. 662-218538
Ms. Nghiem Thi Tu Hien	National Institute for Soil and Fertilizer	Tel. 8404-83856

	Dong Ngac, Tu Liem, Hanoi, Vietnam	Fax 8404-83856
Mr. Oliver B. Barbosa	University of the Philippines, Diliman, Quezon City, the Philippines	Tel. 63-91775861
Mr. Panomsak Promburom	Multiple Cropping Center, Chiang Mai University 239 Huay Kaew Road, Chiang Mai 50200, Thailand	Tel. 6653-221275 Fax. 6653-21000
Mr. Paolo C. Campo	GEODETIC Eng'G Department University of the Philippines, Diliman, Quezon City, the Philippines	Tel. 63-91752888
Dr. Philippe Guizol	Jalan CIFOR Situ Gede, Sindang Barang Bogor Barat 16680, Indonesia	Tel. 62251-62262 Fax 62251-62210
Ms. Princess Alma B. Ani	EcoInformatics Lab, IBS, UPLB College Laguna 4031, the Philippines	Tel. 63-4953674
Dr. Rhodora M. Gonzalez	P.O. Box 346, U.P. Campus Post Office Diliman, Quezon City 1101, the Philippines	Tel. 632-9318320 Fax. 632-434363
Mr. Sombat Yumuang	Geology Department, Faculty of Science Chulalongkorn University, Bangkok 10330, Thailand	Tel. 662-2185442 Fax. 662-218546
Mr. Tayan Raj Gurung	DRDS, Ministry of Agriculture, Thimphu, Bhutan AGS Program, Multiple Cropping Center Chiang Mai University, Chiang Mai 50200, Thailand	Tel. 975-2322383 Tel. 6653-221275 Fax. 6653-21000
Mr. Thinley Gyamtsho	RNRRC-BAJO, WANDUE DRDS, Ministry of Agriculture, Thimphu, Bhutan	Tel. 975-2481209 Fax 975-2481311
Mr. Wahyu Whardana	Faculty of Forestry, Gadjahmada University Balaksumur Yogyakarta 55281, Indonesia	Tel. 622-7451881
OBSERVERS		
Dr. John S. Caldwell	Japan International Research Center for Agricultural Sciences,1-1 Ohwashi, Tsukuba, Ibaraki 3058686, Japan	Tel. 81-29838634 Fax 81-29838634
Mr. Sk. Morshed Anwar	Mail Box 1328, AIT P.O. Box 4 Klong Luang, Pathum Thani 12120	Tel. 662-5248081
Mr. Nicolas Becu	Sam Lan Soi 6 Amphur Muang Chiang Mai 50200, Thailand	Tel. 6653-874487
Mr. Pascal Perez	RMAP Program, RSPAS Australian National University, Australia	Tel. 61-26125810
Mr. Carsten Riedel	Hohenheim University and Uplands Project at CMU Chiang Mai University, Chiang Mai 50200, Thailand	
Mr. Somsak Sukchan	Soil Survey Laboratory, ITCAD, Mitrapap Road Amphoe Muang Khon Kaen 40000, Thailand	Tel. 6643-246758

Appendix 4. Results of the course evaluation by the participants.

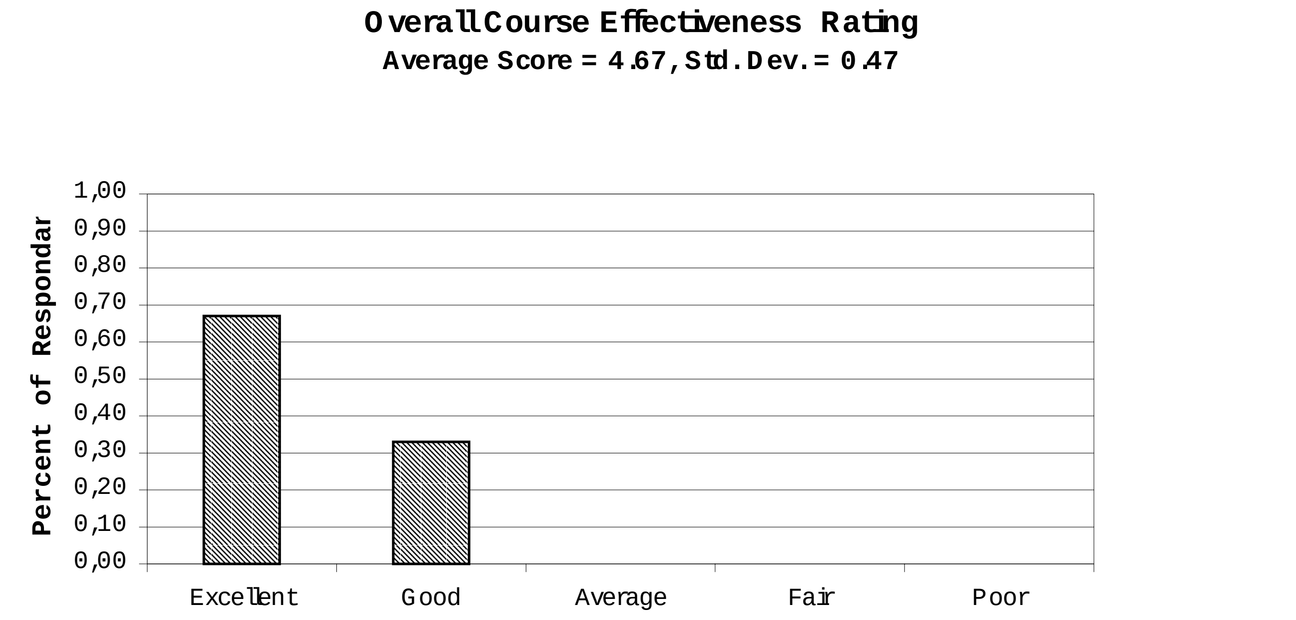


Figure 1.

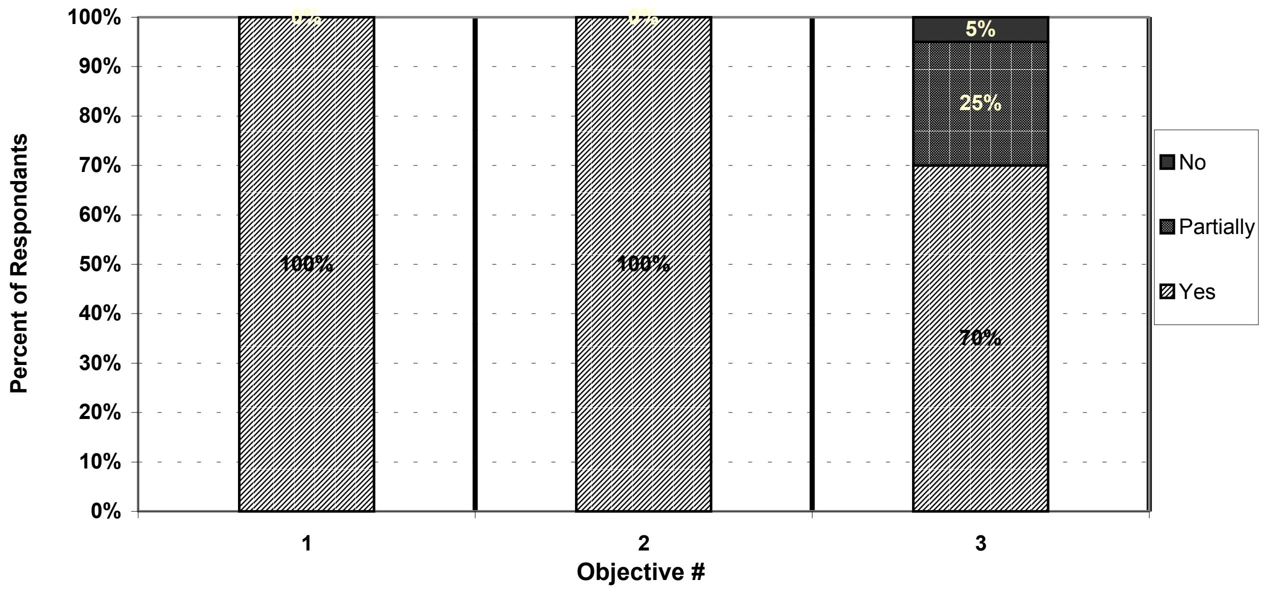
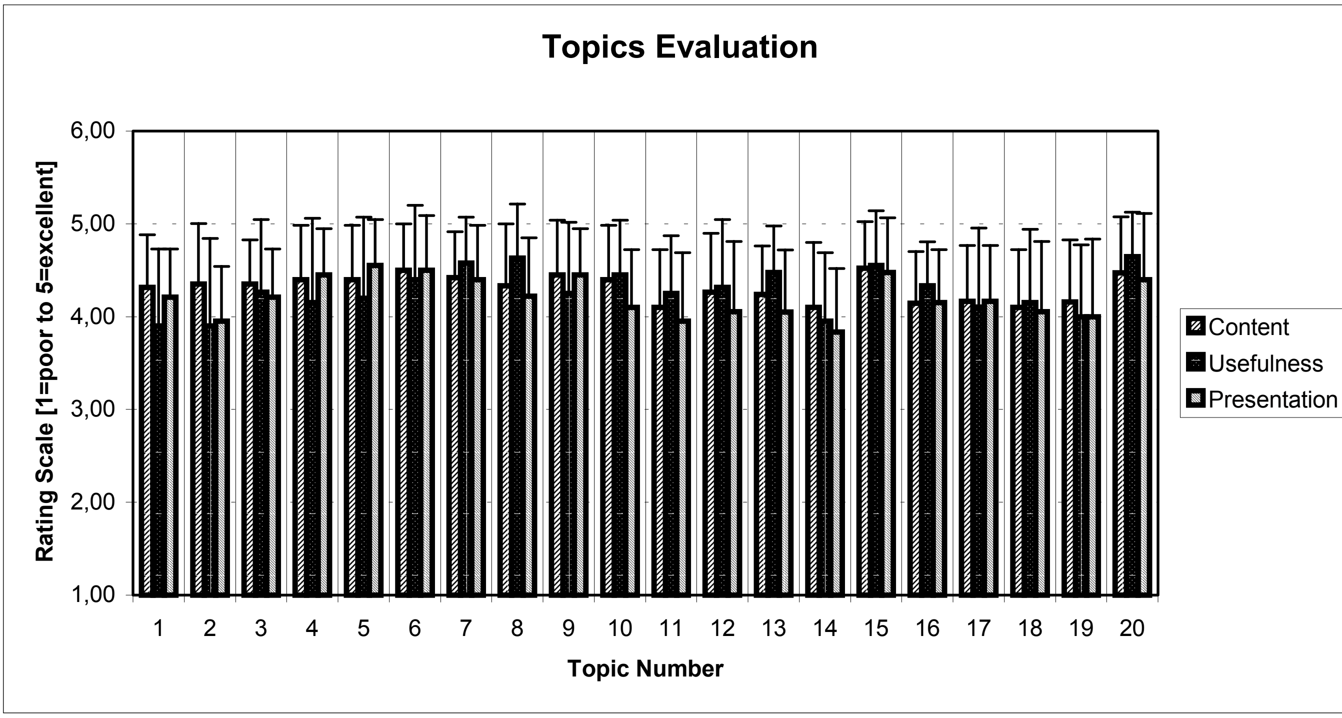


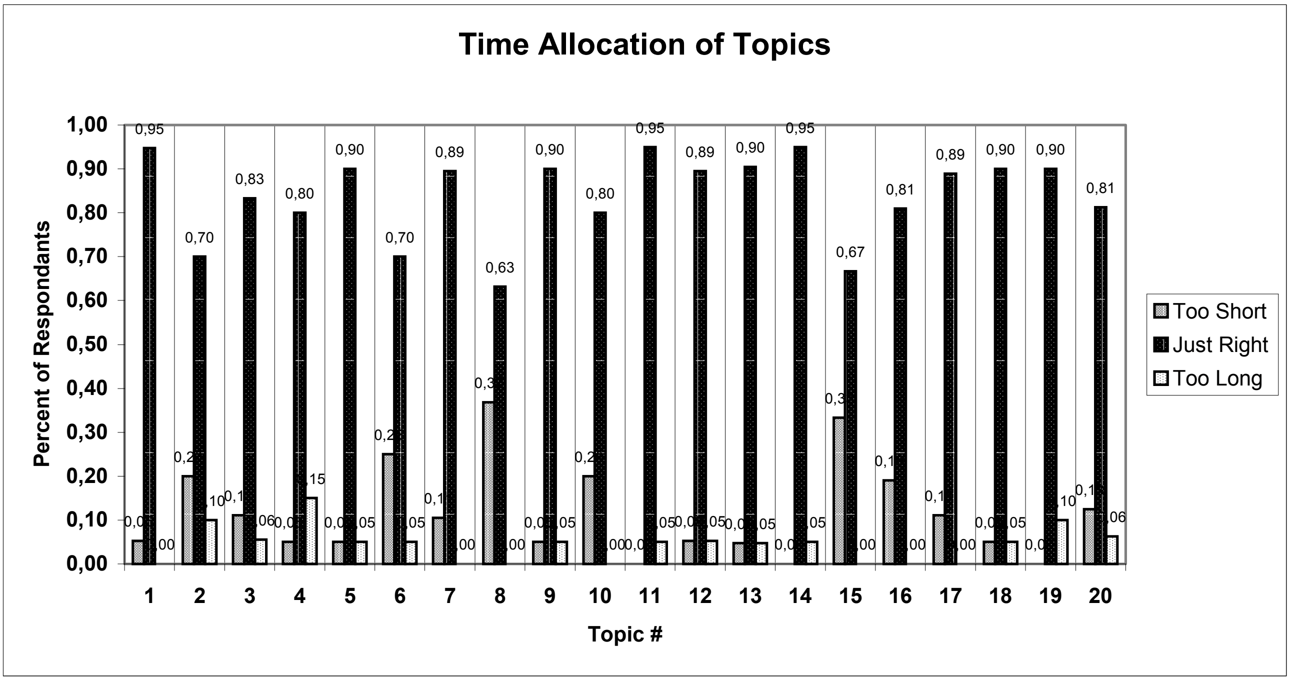
Figure 2. Trainee Evaluation of Attainment of Course Objectives (#s 1 - 3)

Course objectives:

1. Improvement in my understanding of the concepts, approaches, and methods using MAS with GIS for integrated watershed management
2. Improvement in my knowledge and skills related to several tools to use MAS with GIS for integrated watershed management
3. Create a simple application or make progress in improving my personal MAS application under development



Nb.	List of topics
1	Intro. Conference / "Mejan" case study
2	Shadoc role-playing game
3	Introduction to MAS
4	Introduction to GIS
5	Exercice on IDRISI 32
6	Conversion of GIS files for MAS
7	Spatialized MAS
8	Exercices on UML language
9	MAS-GIS-IWM: "Mae Salaep" case study
10	Exercices on linking MAS and GIS
11	Intro. to integrated watershed managt
12	Water management: "Bac Lieu" case
13	Tools for integrated watershed managt
14	Camargue / "Butor Star" case study
15	"Mae Salaep" role-playing game
16	MAS and decision-making processes
17	Exercices on CORMAS: Fire(men)
18	IWM case study: Drome river
19	Case study: the FIRMA Project
20	Group work on personal projects



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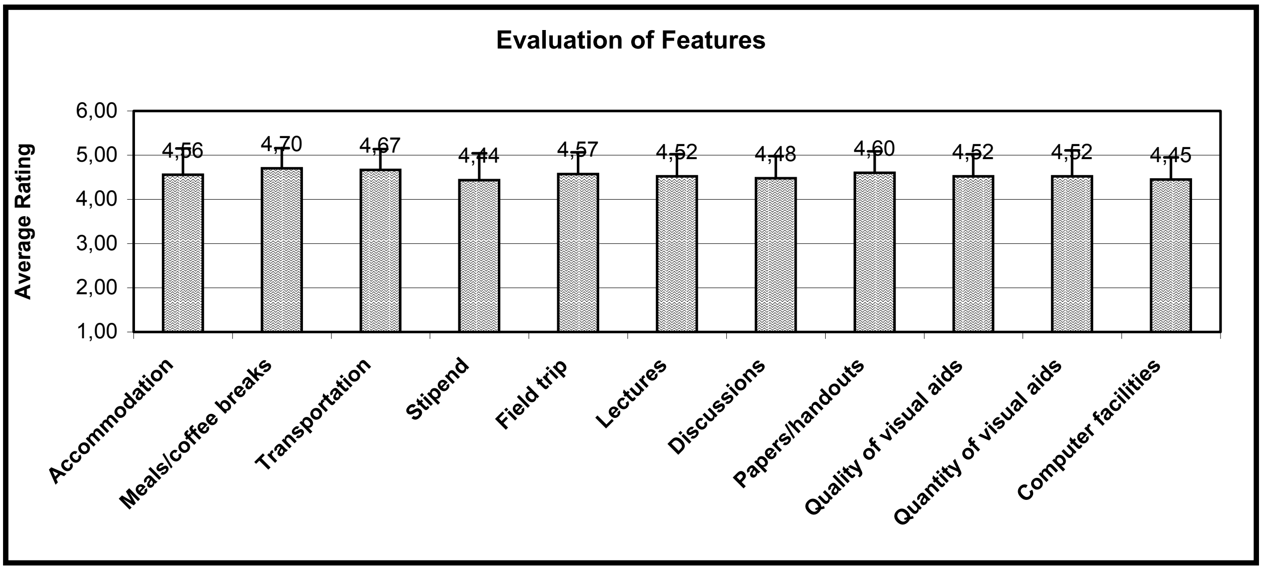
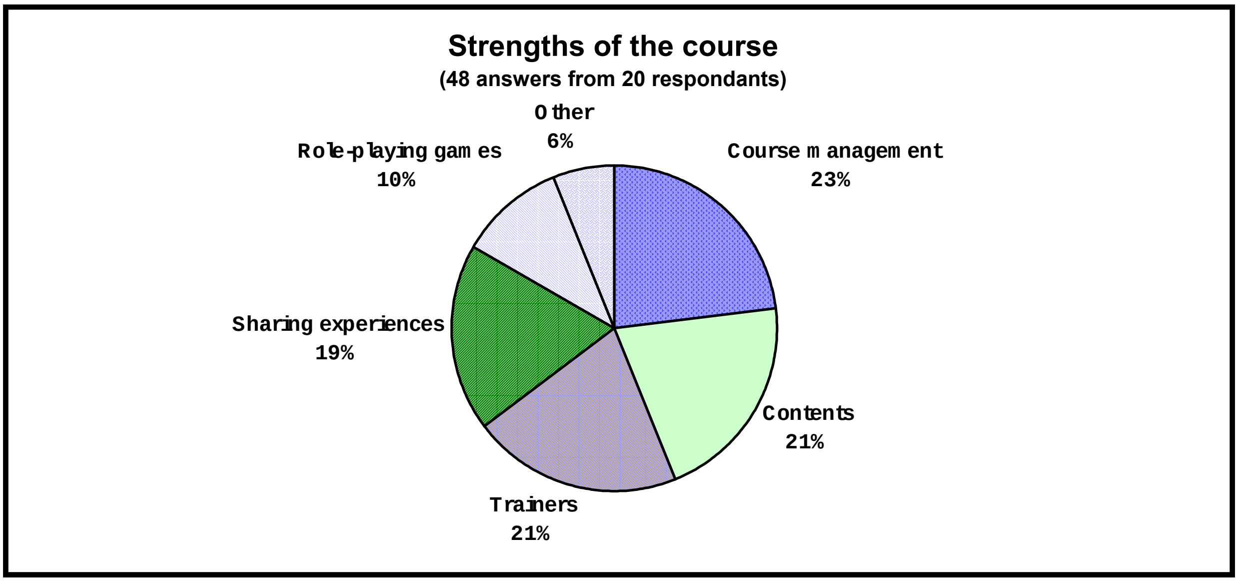
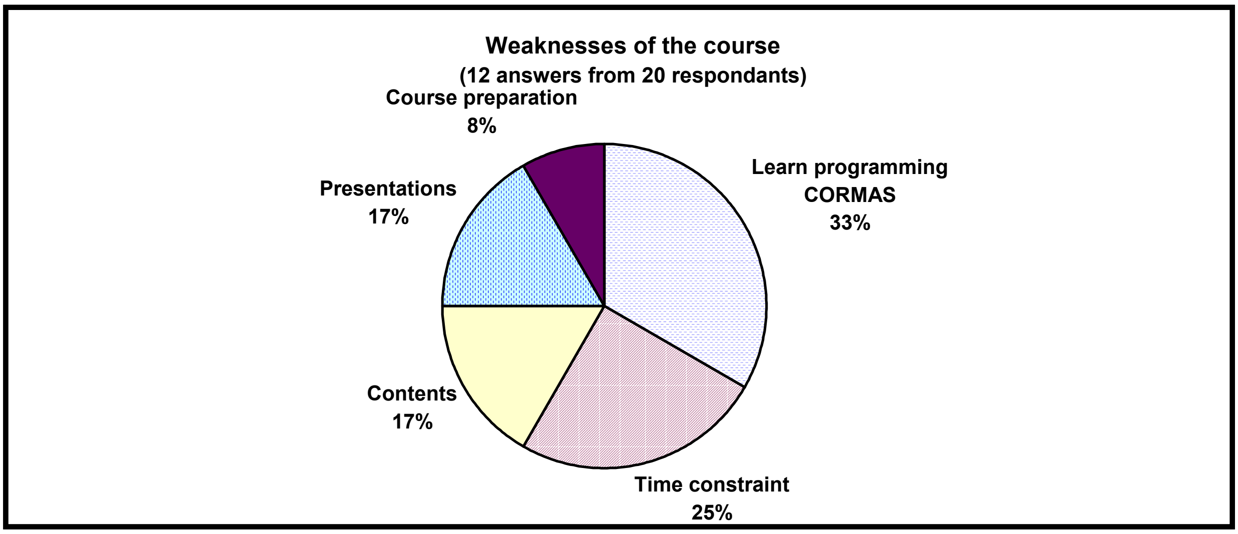

Evaluation of Features
Average Rating
6,00
5,00
4,00
3,00
2,00
1,00
4,56
4,70
4,67
4,44
4,57
4,52
4,48
4,60
4,52
4,52
4,45
Accommodation
Meals/coffee breaks
Transportation
Stipend
Field trip
Lectures
Discussions
Papers/handouts
Quality of visual aids
Quantity of visual aids
Computer facilities


Strengths of the course
(48 answers from 20 respondants)
Other
6%
Role-playing games
10%
Course management
23%
Sharing experiences
19%
Contents
21%
Trainers
21%


Weaknesses of the course
(12 answers from 20 respondants)
Course preparation
8%
Presentations
17%
Learn programming
CORMAS
33%
Contents
17%
Time constraint
25%